



Probing the influence of nonuniform Pt particle size distribution using a full three-dimensional, multiscale, multiphase polymer electrolyte membrane fuel cell model



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ARTICLE INFO

Article history:

Received 21 September 2021

Revised 21 November 2021

Accepted 29 December 2021

Available online 31 December 2021

Keywords:

Fuel cell catalyst layer

Pt particle size distribution

Probability density function (PDF)

Numerical simulation

ABSTRACT

Platinum (Pt) nanoparticles are used as an electrocatalyst in polymer electrolyte membrane (PEM) fuel cells. Visualization data of PEM fuel cell catalysts have shown that the size of Pt particles in the catalyst layer (CL) varies locally from several nanometers to tens of nanometers, owing to the agglomeration and degradation of Pt particles. Although many computational-fluid-dynamics-based PEM fuel cell modeling and simulation studies have been conducted, the majority of them have used the average value of the size of Pt particles owing to the complexity involved in considering the nonuniformity of the Pt particle size in the CL. In this study, we present a new approach to fuel cell CL modeling in PEM fuel cell simulations; the approach involves the consideration of various Pt particle size distributions. In the approach, a probability density function is used to randomly distribute Pt particles with different sizes in a computational domain under the assumption that the Pt particle size distribution follows a normal distribution curve. A sufficient number of grids, typically more than several million grid points, are used to obtain a Pt particle size distribution similar to that in a real CL. The new approach was applied to a comprehensive multiscale PEM fuel cell model and full three-dimensional fuel cell simulations were performed for different CL designs, Pt catalyst degradation levels, and operating conditions. Numerical simulation results clearly showed the significant effect of a nonuniform Pt particle size distribution on multidimensional contours of species concentration, temperature, and current density as well as the overall cell performance.

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1. Introduction

Owing to the increasing global population and increasing industrial energy demand, an enormous amount of qualitative research has been devoted to seeking alternative energy systems [1–3]. Proton exchange membrane (PEM) fuel cells convert directly chemical energy into electrical energy through electrochemical reactions occurring in their catalyst layers (CLs) [4–6], and they are one of the most popular fuel cells because of their attractive features such as high current density, flexibility (in the context of their installation and operation), and low operating temperature. The CL has a very complex and infinitesimal pore structure where three phases (a platinum/carbon (Pt/C) solid phase, the electrolyte phase, and a void phase) randomly occur, apart from a nanoscale pore size distribution and heterogeneous wetting characteristics (hydrophobic and hydrophilic pores). These features render the accurate determination of the electrochemical behavior of and mass transport in

CLs difficult [7,8]. In particular, the electrode material for a PEM fuel cell (PEMFC) should be chosen by considering its reliability, cost, and durability [9,10]. For the maximization of the active Pt surface area in a PEMFC, the size of Pt particles, the Pt loading content, and the Pt particle distribution are crucial factors [11,12].

Recently, Jahnke et al. [13] developed a 2D PEMFC performance model coupled with a complete degradation model to study the effects of cathode catalyst layer degradation. They modeled the aging phenomena of Pt particles, such as oxidation, dissolution, Ostwald ripening, and band formation near the membrane and found that the Pt dissolution rate strongly depends on the size of Pt particles; the smaller the particles, the faster the dissolution [14]. Furthermore, the uniformity of Pt distribution can vary, depending on the catalyst loading and carbon support morphology. Notably, poor catalyst particle dispersion and agglomeration in as-prepared catalyst powders caused rapid coarsening in the early stages of operation. Electrochemical active surface area (ECSA) degradation of catalysts occurs through coarsening resulting from particle coalescence and dissolution [15]. Several studies have investigated the effect of Pt particle size on oxygen reduction reaction (ORR) activity, and the typical optimal Pt particle size for the ORR has been reported to be

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Nomenclature

a	ratio of active surface area to unit electrode volume, m^2/m^3
a_{Pt}	active volumetric surface area of Pt, m^2/m^3
a_{ECSA}	electrochemical active surface area of Pt, $\text{m}^2/\text{g}_{\text{Pt}}$
a_c	volumetric surface area of water, m^2/m^3
A	area, m^2
C	molar concentration of species, mol/m^3
D	species diffusivity, m^2/s
E	activation energy, kJ/mol
EW	equivalent weight of a dry membrane, kg/mol
F	Faraday's constant, $96,487 \text{ C}\cdot\text{mol}^{-1}$
i_0	exchange current density, A/cm^2
I	operating current density, A/cm^2
I/C	ionomer-to-carbon weight ratio
j	transfer current density, A/cm^2
k	thermal conductivity, $\text{W}/(\text{m}\cdot\text{K})$
K	hydraulic permeability, m^2
L_{Pt}	amount of Pt loading, mg/cm^2
n	number of electrons transferred in the electrode reaction
n_{Pt}	number of Pt particles on a Pt/C particle
MW	molecular weight, kg/mol
P	pressure, Pa
r	particle radius, m
R	universal gas constant, $8.314 \text{ J}/(\text{mol}\cdot\text{K})$
s	liquid saturation
S	source term in the transport equation
T	temperature, K
$\bar{u}$	fluid velocity and superficial velocity in a porous medium, m/s
z	transport resistance coefficient

Greek symbols

α	transfer coefficient
γ	reaction order
δ	thickness, m
ε	volume fraction
η	surface overpotential, V
κ	proton conductivity, $\text{S}\cdot\text{m}^{-1}$
ϕ	phase potential, V
ρ	density, kg/m^3
σ	electronic conductivity, S/m
τ	viscous shear stress, N/m^2
ξ	stoichiometry flow ratio
Ω	oxygen transport resistance

Superscripts

c	catalyst coverage coefficient
eff	effective
g	gas
l	liquid
Pt	platinum
TiO_2	titanium oxide
ref	reference value

Subscripts

an	anode
ca	cathode
c	carbon
CL	catalyst layer
e	electrolyte
GC	gas channel
GDL	gas diffusion layer

H_2	hydrogen
i	species
in	channel inlet
mem	membrane
O_2	oxygen
s	solid, surface
T	temperature
u	momentum equation
w	water
0	initial conditions or standard conditions (i.e., 298.15 K and 101.3 kPa (1 atm))

in the range of 2–4 nm [16–18]. However, during long-term PEMFC operation, it is challenging to maintain the Pt particle size in this range. Agglomeration and growth of Pt particles have been widely observed in transmission electron microscopy (TEM) images [19–22]. For instance, Mohamed et al. [23] reported that Pt particle sizes in a Pt/C electrocatalyst-based CL before and after the durability test for 10,000 cycles were 1–6 and 2–14 nm, respectively, which resulted in a 50% loss of the initial cell potential. In another study, Zhang et al. [24] investigated the stability of a Pt/C catalyst by potential cycling between 0.6 and 1.2 V. After 4000 cycles, the initial ECSA decreased considerably, from $61.93 \text{ m}^2/\text{g}_{\text{Pt}}$ to $33.74 \text{ m}^2/\text{g}_{\text{Pt}}$, while the Pt particle size increased from the range of 1–6 nm to 1–25 nm.

Substantial efforts have been made to numerically investigate the effect of Pt particle size on electrochemical activity and cell performance. The majority of numerical works have assumed Pt particles to have a constant and uniform size [25,26]. In one study, an agglomerate model in which a large number of Pt/C catalyst particles were homogeneously mixed with an ionomer was simulated [27]. In the simulations, the average radii of agglomerate and Pt particles were assumed to be constant at 1000 and 1.25 nm, respectively, and the particles were distributed in a $15 \mu\text{m}$ thick CL. The simulation results showed that the reaction rate within the CL varied significantly with location. Recently, Zhang et al. [28] developed a three-dimensional multiphase numerical model to accurately predict the effect of agglomeration of spherical Pt particles on the concentration loss in PEMFCs. They found that the incorporation of the agglomerate model in the numerical model facilitated a detailed investigation of the effect of the CL composition and Pt loading on the microscopic heat and mass transfer in PEMFC. Carcadea et al. [29] numerically analyzed the effect of Pt particle radius on a 5 cm^2 PEMFC's performance. They showed that (around 25%) the fuel cell's performance was significantly degraded with an increase in the radius of the Pt particles from 1.61 nm to 5.5 nm. Another study combined the pore-scale (PS) model with the volume-averaged (VA) model and analyzed the spatial heterogeneity of a gas diffusion layer (GDL) using the PS model. It found that the VA model cannot accurately predict degradation, durability, and the interface behavior [30].

The aforementioned experimental studies involving TEM analysis have shown that the growth of Pt nanoparticles is uneven in CLs, resulting in a wide range of Pt particle size distributions, from several nanometers to tens of nanometers. However, numerical studies to account for the effects of a nonuniform Pt particle size distribution have not been reported, mainly because of inherent difficulties in modeling and simulating electrochemical and transport processes in a complex and infinitesimal CL pore structure. The objective of this study was to numerically elucidate the effects of Pt particle growth and a nonuniform Pt particle size distribution on detailed distributions of species and current density as well as the overall cell performance. A multiscale, multidimensional, multiphase PEMFC developed in our previous works [31–33] was em-

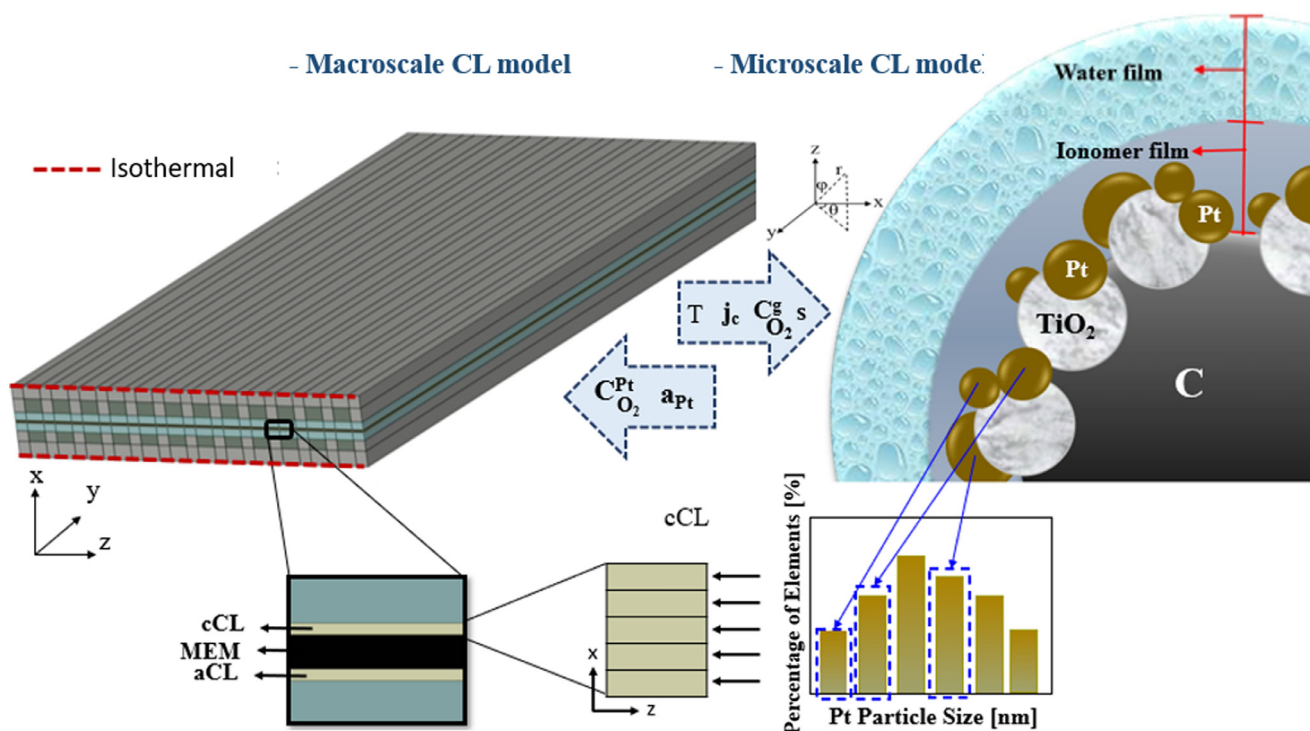


Fig. 1. Micro and Macroscale computational domains of a PEMFC combined with non-uniform distribution of Pt particles using a PDF.

employed. In particular, a microscale CL model based on the spherical coordinate system (r, θ, ϕ) was coupled with a comprehensive electrochemical and transport fully coupled PEMFC model constructed in the Cartesian coordinate system (x, y, z) . For the random distribution of Pt particles of different sizes in a given computational CL domain, a statistical approach involving a probability density function (PDF) was used under the assumption that the Pt particle size distribution followed a normal distribution curve. Multiscale simulations were performed using the Pt particle size distribution data of Liu et al. [34], who conducted the accelerated aging test for two catalyst materials (Pt/C and Pt/TiO₂/C). The simulation results are first compared with experimentally measured polarization curves and then analyzed in detail to elucidate significant effects of Pt particle growth and a nonuniform Pt particle size distribution on distributions of species and current density as well as the overall cell performance.

2. Numerical model

In a previous study [32], a microscale CL model that is suitable for the precise investigation of the effects of the structure and properties of a microscale CL was developed and coupled with a macroscale 3D PEMFC model. Subsequently, the microscale CL model was improved by considering electron transport through a microscale CL structure, which is particularly important for CL designs in which semi-conductive metal materials (e.g., TiO₂) are used as a support for Pt nanoparticles to resolve degradation issues occurring in typical Pt/C based CLs [33]. Although multiscale simulations can successfully indicate the effects of microscale CL properties and structural designs, one of the differences from an actual situation is that Pt particles of the same size are uniformly distributed on the surface of a carbon support in the microscale CL domain. To determine the effects of a nonuniform Pt particle distribution in CLs on the basis of experimental TEM images of CLs, we used a PDF in the CL domain wherein Pt particles with differ-

ent sizes were randomly distributed. This statistical approach ensures that local Pt particle size varies from several nanometers to tens of nanometers, and a schematic of the application methodology is shown in Fig. 1. Accordingly, the 3D PEMFC model combined with the PDF approach can capture more realistic behavior of the electrochemical reaction and oxygen and charge transport, which is strongly affected by local Pt particle size. The computational domain included various sizes of Pt particles and carbon particles covered by an ionomer film and water film as well as subregions of the PEMFC, namely, gas channels (GCs), GDLs, CLs, a membrane, and bipolar plates (BPs). The following were the primary assumptions made for the evaluation of the above system model.

- Pt particles are spherical and have a wide range of sizes.
- An ionomer film surrounds the Pt particles.
- Owing to typical low operating pressure for fuel cells, gas mixtures follow the ideal gas laws.
- Gravitational effects are insignificant.
- Gas flow is incompressible and laminar.
- The effect of immobile liquid saturation in the porous CL and GDLs can be neglected. This is reasonable for a high level of fuel cell flooding.

2.1. Two-phase PEMFC model on 3D macroscale

The multi-scale, two-phase PEMFC model is governed by five conservation principles: mass, momentum, species, charge, and thermal energy. The conservation equations were coupled with source terms representing electrochemical reactions, namely, hydrogen oxidation reaction (HOR) at the anode and ORR at the cathode, and the kinetic expressions for the ORR and HOR, given in Eqs. (8) and (9), respectively, were derived using the standard Butler-Volmer equation. All the governing equations and source terms are listed in Tables 1 and 2, respectively, and the physiochemical, kinetic, and transport properties are presented in Table 3.

Table 1
A 3D two-phase PEMFC model: Governing equations

Governing equations		
Mass	$\nabla \cdot (\rho \vec{u}) = 0$	(1)
Momentum	$\left(\frac{1}{\varepsilon^2}\right) \nabla \cdot (\rho \vec{u} \vec{u}) = -\nabla P + \nabla \cdot \tau + S_u$	(2)
Species	Flow channels and porous media: $\nabla \cdot (\gamma_i \rho m_i \vec{u}) = \nabla \cdot [\rho^g D_i^{g,eff} \nabla (m_i^g)] + \nabla \cdot [(m_i^g - m_i^l) \vec{j}] + S_i$	(3)
	Water transport in the membrane: $\nabla \cdot \left(\left(\frac{\rho^{mem}}{EW} \right) D_w^{mem} \nabla \lambda \right) M_w - \nabla \cdot \left(n_d \left(\frac{l}{F} \right) \right) M_w + \nabla \cdot \left(\left(\frac{\kappa^{mem}}{v^l} \right) \nabla P^l \right) = 0$	(4)
Charge	Proton transport: $\nabla \cdot (\kappa^{eff} \nabla \phi_e) + S_\phi = 0$	(5)
	Electron transport: $\nabla \cdot (\sigma^{eff} \nabla \phi_s) - S_\phi = 0$	(6)
Energy	$\nabla \cdot (\rho \vec{u} C_p^g T) = \nabla \cdot (k^{eff} \nabla T) + S_T$	(7)

Table 2
The 3D two-phase PEMFC model: source/sink terms.

Source/sink terms		
Momentum	For porous media:	$S_u = -\frac{\mu}{K} \vec{u}$
Species	For water in the CLs:	$S_i = \left[-\nabla \cdot \left(\frac{n_d}{F} \right) - \frac{s_{ij}}{nF} \right] M_i$
	For other species in the CLs:	$S_i = -M_i \frac{s_{ij}}{nF}$
Energy	In the CLs:	$S_T = j \left(\eta + T \frac{dU_0}{dT} \right) + \frac{l_e^2}{k^{eff}}$
	In the membrane:	$S_T = \frac{l_e^2}{k^{eff}}$
Charge	In the anode CL:	$S_{\phi,an} = j_{an}$
	In the cathode CL:	$S_{\phi,ca} = j_{ca}$
Electrochemical reactions		
$\sum_k s_i M_i^z = ne^-$, where $\begin{cases} M_i = \text{chemical formula of species } i \\ s_i = \text{stoichiometry coefficient} \\ n = \text{number of electrons transferred} \end{cases}$		
Hydrogen oxidation reaction (HOR) at the anode side: $H_2 - 2H^+ = 2e^-$		
Oxygen reduction reaction (ORR) at the cathode side: $2H_2O - O_2 - 4H^+ = 4e^-$		
Transfer current density, $[A/m^2]$	HOR in the anode CL:	(8)
	$j = (1-s) a_{i,an}^{ref} \left(\frac{C_{H_2}}{C_{H_2,ref}} \right)^{\frac{1}{2}} \exp \left[-\frac{E_{an}}{R} \left(\frac{1}{T} - \frac{1}{353.15} \right) \right] \left(\exp \left(\frac{\alpha_{an} F}{RT} \eta \right) - \exp \left(-\frac{\alpha_{ca} F}{RT} \eta \right) \right)$	
	ORR in the cathode CL:	(9)
	$j = -\frac{3L_{Pt}}{r_{Pt} \cdot \rho_{Pt} \cdot \delta_{CL}} i_{0,ca}^{ref} \left(\frac{C_{O_2}^{Pt}}{C_{O_2,ref}} \right)^{3/4} \exp \left(-\frac{E_{ca}}{R} \left(\frac{1}{T} - \frac{1}{353.15} \right) \right) \exp \left(-\frac{\alpha_{ca}}{R_u T} F \eta \right)$	
Overpotential	$\eta = \phi_s - \phi_e - U$	(10)
	where, $U = U_0 - \frac{RT}{nF} \ln \frac{C_{O_2}}{C_{O_2,ref}}$	

2.2. Microscale CL model

The parameters of a CL particle in a microscale CL domain that appear model equations are classified in Table 4, and they include the volume fractions of Pt (ε_{Pt}), carbon (ε_c), ionomer (ε_e), and TiO₂ (ε_{TiO_2}), as well as the CL thickness (δ_{CL}); these are based on the ionomer-to-carbon weight ratio (I/C), weight percent of Pt/C and TiO₂/C ($wt\%_{Pt/C}$ and $wt\%_{Pt/TiO_2}$), and Pt loading function (L_{Pt}). Since the ionomer layer is hydrophilic, it is covered with a layer of water, and Eqs. (16) and (17) determine the thickness of the ionomer film and water film, respectively. Furthermore, the oxygen flux across the water and ionomer layers, the volumetric surface area of Pt (a_{Pt}), and a_{ECSA} were formulated using Eqs. (19)–(22), respectively. The variable Ω_T represents the total oxygen transport resistance, which was obtained by using several interfacial resistances, including the interfacial resistance in the ionomer and water films and the effective thicknesses of the water and ionomer films (δ_w^{eff} , δ_e^{eff}). Additional details of “microscale modeling” can be obtained from Refs. [32,35,36,37]. Owing to the presence of TiO₂ particles on some carbon surfaces in the CL domain at the microsphere scale, an additional voltage drop associated with electron transfer through TiO₂ particles can be considered in the microscale

CL model. Expressions for the electron transport resistance across TiO₂ particles, Ω_{TiO_2} , the actual electron transport path across a single TiO₂ particle, $\delta_{TiO_2}^{eff}$, and the additional voltage drop over the TiO₂ particles, $\Delta\phi_s$, are presented in Table 5, in which the expression for j_{ca} was obtained from Eq. (20). The microscale CL model received the local value computed by the 3D macroscale PEMFC model for two million grid points in the computational domain.

2.3. PDF of PEMFC

This section details the methodology for introducing a PDF to our previously developed multiscale model. The characteristics of a PDF are as follows.

- The PDF, expressed as $f(x)$, is non-negative for all x .
- The integral of the PDF over the entire space is always equal to one.
- As the number of samples increases, the reliability increases.

If a PDF is to be considered, a key requirement for a CL is that the Pt particle size distribution should be nonuniform. Moreover, since the total share of Pt particle sizes that equals 1 comprises the share of every Pt particle size, the share can be considered as a probability (P) given by

Table 3
Kinetic, physiochemical, and transport properties.

Description	Value	Ref.
Exchange current density $\times$ ratio of the reaction, $aI_{0,an}^{ref}$	1.2×10^{10} A/m ³	[32]
Reference exchange current density of oxygen reduction reaction, $I_{0,ca}^{ref}$	3.0×10^{-5} A/cm ² -Pt	[32]
Activation energy for anode, E_{an}	10.0 kJ/mol	[36]
Activation energy for cathode, E_{ca}	70.0 kJ/mol	[36]
Transfer coefficient of HOR, $\alpha_{an} = \alpha_{ca}$	1	[36]
Transfer coefficient of ORR, α_{ca}	1	[36]
Reaction order for anode/cathode, γ_{an}/γ_{ca}	0.5/0.75	[32]
Reference H ₂ /O ₂ molar concentration, C^{ref}	40.88 mol/m ³	[36]
Permeability of the GDL/CL, K_{GDL}/K_{CL}	$1.0 \times 10^{-12} / 1.0 \times 10^{-13}$ m ²	[38]
Equivalent weight of electrolyte in membrane, EW	1.1 kg/mol	[39]
Dry membrane density, $\rho_{dry, mem}$	2000 kg/m ³	[39]
Faraday constant, F	96,487 C/mol	
Universal gas constant, R_u	8.314 J/(mol•K)	
H ₂ diffusivity in the anode gas channel, $D_{0,H_2,an}^g$	1.1028×10^{-4} m ² /s	[39]
H ₂ O diffusivity in the anode gas channel, $D_{0,H_2O,an}^g$	1.1028×10^{-4} m ² /s	[39]
O ₂ diffusivity in the cathode gas channel, $D_{0,O_2,ca}^g$	3.2348×10^{-4} m ² /s	[39]
H ₂ O diffusivity in the cathode gas channel, $D_{0,H_2O,ca}^g$	7.35×10^{-5} m ² /s	[39]
O ₂ diffusivities through the ionomer film and water film, $D_{O_2,e}, D_{O_2,w}$	0.58×10^{-10} m ² /s	[40]
Effective electronic conductivity in the CLs, σ_{CL}	1000 S/m	[41]
Effective electronic conductivity in the GDLs, σ_{GDL}	10000 S/m	[41]
Effective electronic conductivity in the BP, σ_{BP}	20000 S/m	[41]
Electronic conductivity in TiO ₂ , σ_{TiO_2}	1.37×10^{-5} S/m	[42]
Density of carbon particle, ρ_c	1.95 g/cm ³	[43]
Density of TiO ₂ particle, ρ_{TiO_2}	4.23 g/cm ³	[44]
Density of platinum, ρ_{Pt}	21.45 g/cm ³	[45]
Density of water, ρ_w	1.0 g/cm ³	[46]
Density of ionomer, ρ_e	1.9 g/cm ³	[32]
Transport resistance coefficient at ionomer film, z_e	6.5	[32]
Transport resistance coefficient at Pt particle, z_{Pt}	5.4	[32]
Transport resistance coefficient at water film, z_w	6.5	[32]

$P(a \leq X \leq b) = \int_a^b f(x)dx$ in the case of a continuous probability distribution Eq. (32)

$P(X = x) = f(x)$ in the case of a discrete probability distribution Eq. (33) where P for a random variable X with a value between a and b can be calculated as the integral of the PDF over the intervals a and b . In general, the Pt particle size distribution is given as a histogram, and it corresponds to a discrete probability distribution, which makes the process of applying a PDF to the CL model easier. Specifically, if the discrete probability distribution is expressed as a cumulative distribution, P of each random variable can be expressed as follows:

$$P(X \leq x_n) = \sum_{n=1}^n f(x_n) \quad (34)$$

where n denotes the n th random variable on the x -axis of the histogram. The cumulative probability corresponding to the final random variable is 1, and it is the sum of P 's corresponding to all the preceding random variables. A schematic of the cumulative probability distribution is shown in Fig. 2. To consider a discrete probability distribution for the Pt particle size, we should stratify and assign a random number in the range from 1 to 100 to each random variable according to the P of the random variable as follows:

$$X_1 = \{ 1, 2, \dots, 100 \times P_1 \}$$

$$X_2 = \{ 100 \times P_1 + 1, 100 \times P_1 + 2, \dots, 100 \times P_2 \}$$

⋮

$$X_n = \{ 100 \times P_{n-1} + 1, 100 \times P_{n-1} + 2, \dots, 99, 100 \}$$

If each random variable is regarded as a Pt particle size distribution in the CL, the Pt particle size can be determined by select-

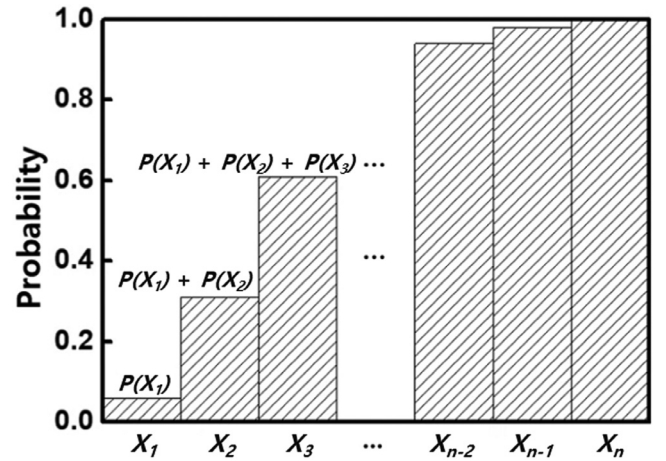


Fig. 2. Schematic of cumulative probability distribution.

ing a random number between 1 and 100. For instance, if a random number belongs to the range $100 \times P_1 + 1 < x < 100 \times P_2$, it will have Pt particle size X_2 . In order to consider a nonuniform Pt particle size distribution in the computational domain, the methodology mentioned above is applied to each element of CL. Furthermore, since P is applied by the sampling with replacement to each element, a previously determined Pt particle size does not affect the Pt particle size of the later elements. In other words, the Pt particle size assigned to an element is determined only by using a stratified random number between 1 and 100 extracted from the element, and this process is performed on all elements individually. Moreover, in order to obtain the reliability of the PDF model, the grid is constructed very finely and the minimum value of element size represents 1 μ m.

Table 4
Correlations of the microscale CL model.

Description	Expression	
Carbon volume fraction (ε_c)	$\varepsilon_c = \frac{L_{Pt}}{\delta_{CL}} \left(\frac{1}{\rho_c \text{wt}\%_{Pt/C}} \right)$	(11)
Pt particle volume fraction (ε_{Pt})	$\varepsilon_{Pt} = \frac{1}{\rho_{Pt}} \frac{L_{Pt}}{\delta_{CL}}$	(12)
Ionomer volume fraction (ε_e)	$\varepsilon_e = \left(\frac{I}{C} \right) \frac{\rho_c}{\rho_e} \varepsilon_c$	(13)
TiO ₂ volume fraction (ε_{TiO_2})	$\varepsilon_{TiO_2} = \frac{L_{Pt}}{\delta_{CL}} \left(\frac{1}{\rho_{TiO_2} \text{wt}\%_{Pt/TiO_2}} \right)$	(14)
CL thickness (δ_{CL}) [m]	$\delta_{CL} = \left[\frac{\left(\frac{I}{C} \right) \frac{\rho_c}{\rho_e} \left(\frac{1}{\rho_c \times \text{wt}\%} \right) + \left(\frac{1}{\rho_{Pt}} + \frac{1}{\rho_c \times \text{wt}\%} \right)}{1 - \varepsilon_{CL}} \right] L_{Pt}$	(15)
Ionomer film thickness (δ_e) [m]	$\delta_e = \left[\left(1 + \frac{\varepsilon_e}{\varepsilon_c} \right)^{\frac{1}{3}} - 1 \right] r_c$	(16)
Water film thickness (δ_w) [m]	$\delta_w = \left[\left(1 + \frac{\varepsilon_e}{\varepsilon_c} + \frac{s}{\varepsilon_c} \right)^{\frac{1}{3}} - 1 \right] r_c - \delta_e$	(17)
Oxygen balance in a single Pt-TiO ₂ /C particle domain [mol/m ³]	$C_{O_2}^g - C_{O_2}^{Pt} = -\Omega_T \frac{j_{ca}}{4F \cdot a_{ca}} = \Omega_T \frac{I}{4F \cdot \delta_{CL} \cdot a_{ca}} = R_T \frac{I}{4F}$	(18)
Volumetric surface area of ionomer (a_c) [m ² /m ³]	$a_c = \frac{3\varepsilon_c}{r_c^3} (r_c + \delta_{TiO_2} + \delta_e + \delta_w)^2$	(19)
Transfer current density [A/m ³]	$j_{ca} = -a_{Pt,ca}^{ref} \left(\frac{C_{O_2}^{Pt}}{C_{O_2,ref}} \right)^{\gamma_{ca}} \exp \left(-\frac{E_{ca}}{R} \left(\frac{1}{T} - \frac{1}{353.15} \right) \right) \exp \left(-\frac{\alpha_{ca}}{RT} F \eta_{ca} \right)$	(20)
Volumetric surface area of Pt (a_{Pt}) [m ² /m ³]	$a_{Pt} = \frac{a_{ECSA} L_{Pt}}{\delta_{CL}}$ $(a_{ECSA} = \frac{3}{r_{Pt} \cdot \rho_{Pt}})$	(21) (22)
Total oxygen transport resistance (Ω_T) [s/m]	$\Omega_T = \Omega_{w,int} + \frac{\delta_w^{eff}}{D_{O_2,w}} + \Omega_{e,int} + \frac{\delta_e^{eff}}{D_{O_2,e}} + \Omega_{Pt,int}$	(23)
Total transport resistance [s/m]	$R_T = \frac{\Omega_T}{\delta_{CL} \cdot a_c}$	(24a)
Interfacial resistance in ionomer [s/m]	$\Omega_{e,int} = z_e \frac{\delta_e}{D_{O_2,e}}$	(24b)
Interfacial resistance in water film [s/m]	$\Omega_{w,int} = z_w \frac{\delta_w}{D_{O_2,w}}$	(24c)
Interfacial resistance in platinum particle [s/m]	$\Omega_{Pt,int} = z_{Pt} \frac{(r_c + \delta_{TiO_2} + \delta_e)^2}{r_{Pt}^2} \frac{\rho_{Pt}}{r_c} \left(\frac{1 - \text{wt}\%}{\text{wt}\%} \right) \frac{\delta_e}{D_{O_2,e}}$	(24d)
Effective ionomer thickness [m]	$\delta_e^{eff} = \frac{(r_c + \delta_{TiO_2} + \delta_e + \delta_w)^2}{n_{Pt} \cdot r_{Pt}^2} \delta_e$	(25)
Effective water thickness [m]	$\delta_w^{eff} = \frac{(r_c + \delta_{TiO_2} + \delta_e + \delta_w)^2}{n_{Pt} \cdot r_{Pt}^2} \delta_w$	(26)
Number of Pt particles on a single carbon particle	$n_{Pt} = \frac{\rho_c}{\rho_{Pt}} \left(\frac{r_c}{r_{Pt}} \right)^3 \left(\frac{\text{wt}\%}{1 - \text{wt}\%} \right)$	(27)
Oxygen concentration on Pt particle surfaces ($C_{O_2}^{Pt}$) [mol/m ³]	$f(C_{O_2}^{Pt}) = B(C_{O_2}^{Pt})^\gamma + \frac{4F}{\Omega_T} \frac{3\varepsilon_c}{r_c^3} (r_c + \delta_e)^2 C_{O_2}^{Pt} - \frac{4F}{\Omega_T} \frac{3\varepsilon_c}{r_c^3} (r_c + \delta_e)^2 C_{O_2}^g = 0$ where $B = \frac{3L_{Pt}}{r_{Pt} \cdot \rho_{Pt} \cdot \delta_{CL}} \left(\frac{1}{C_{O_2,ref}} \right)^{\gamma_{ca}} \exp \left(-\frac{E_{ca}}{R} \left(\frac{1}{T} - \frac{1}{353.15} \right) \right) \exp \left(-\frac{\alpha_{ca}}{RT} F \eta_{ca} \right)$	(28)

Table 5
Correlations of additional microscale CL properties associated with TiO₂.

Description	Expression	
Electron transport resistance [m ² /S]	$\Omega_{TiO_2} = \frac{\delta_{TiO_2}^{eff}}{\sigma_{TiO_2}}$	(29)
Effective TiO ₂ thickness [m]	$\delta_{TiO_2}^{eff} = 2r_{TiO_2} \frac{\rho_{Pt}}{\rho_{TiO_2}} \times \left(\frac{r_{Pt}}{r_{TiO_2}} \right)^3 \times \left(\frac{1 - \text{wt}\%_{TiO_2/TiO_2+Pt}}{\text{wt}\%_{TiO_2/TiO_2+Pt}} \right) \times \left(\frac{1 - \text{wt}\%_{Pt/Pt+C}}{\text{wt}\%_{Pt/Pt+C}} \right)$	(30)
Additional voltage drop [V]	$\Delta \phi_s = \Omega_{TiO_2} \frac{r_c - I}{3\varepsilon_c \cdot \delta_{CL}} =$	(31)

2.4. Boundary conditions and numerical implementation

Constant temperature conditions were applied on each side of the outer BP surfaces, and symmetric conditions were applied to the bottom and top surfaces. Fig. 1 shows the computational domain, mesh structure, and boundary conditions. Table 6 presents detailed cell dimensions and operating conditions, and Table 7 lists the thermal properties. Equations for the boundary conditions are presented in Table 8, where the stoichiometries of the anode and

Table 6
Cell dimensions and operating conditions.

Description	Value
Channel width	0.5 mm
Channel depth (anode/cathode)	0.4/0.6 mm
Rib width	0.3 mm
GDL thickness, δ_{GDL}	0.2 mm
Membrane thickness, δ_{mem}	0.02 mm
Cell length	50 mm
Pressure (anode/cathode), P_{an}/P_{ca}	2/2 bar
Operating temperature, T_{cell}	353.15 K
Stoichiometry (anode/cathode), ξ_{an}/ξ_{ca}	1.2/2.0
Weight percentage of Pt/C, wt%	40%/60%
Weight percentage of C/TiO ₂ , wt%	58.92%/1.08%
Weight percentage of C/TiO ₂ +Pt, wt%	58.92%/41.08%

cathode (ξ_{an} and ξ_{ca}) were used to determine the anode and cathode velocities ($u_{in, an}$ and $u_{in, ca}$). Furthermore, the electric potential on the anode's outer BP surface was zero (Eq. (37)), while that on the cathode's outer BP surface was equal to the current density or cell voltage, whichever was constant (Eq. (38)).

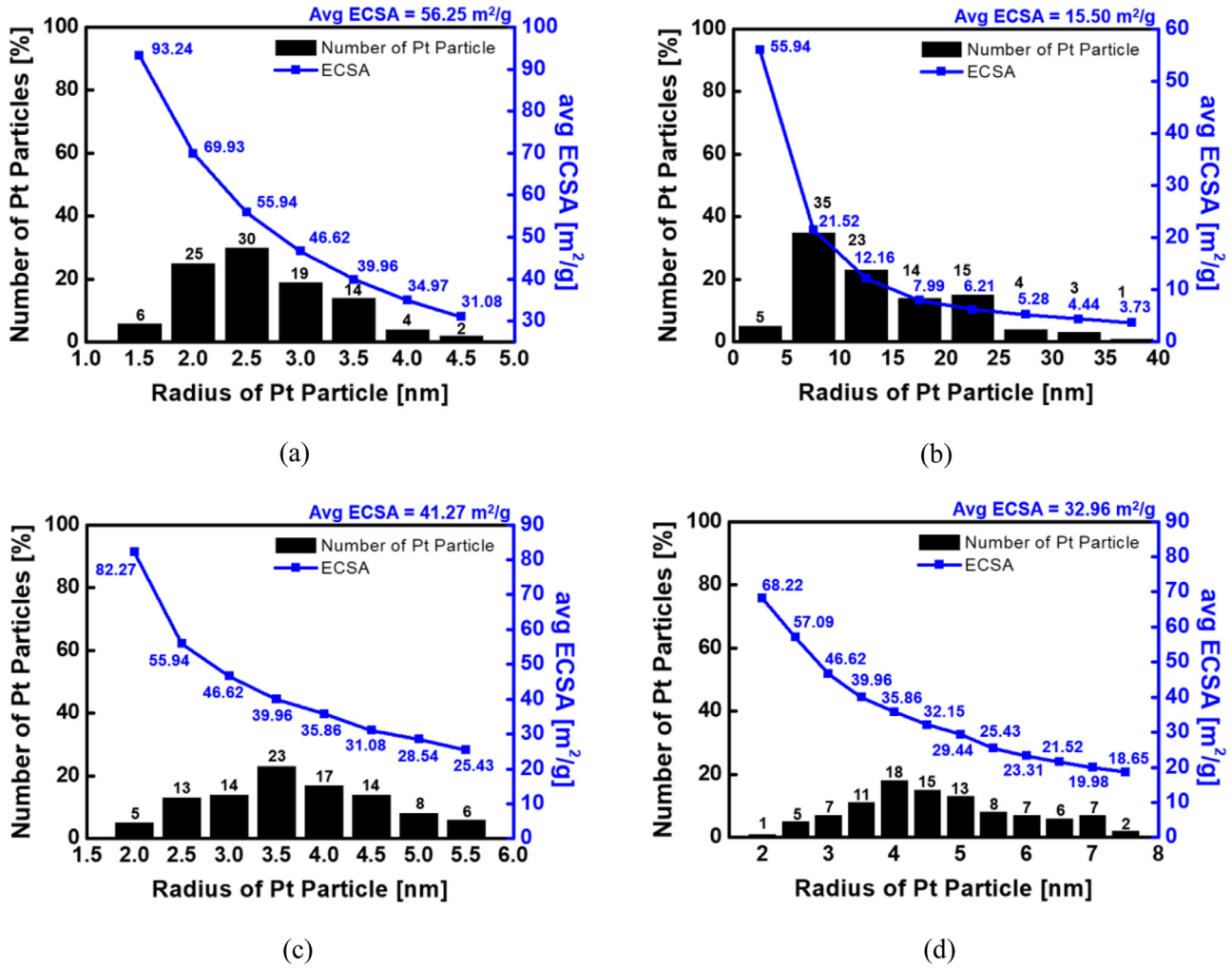


Fig. 3. Histograms of Pt particle size distributions in the cathode CL along with the ECSA. (a) Pt/C catalyst before AAT, (b) Pt/C catalyst after AAT, (c) Pt/TiO₂/C catalyst before AAT, and (d) Pt/TiO₂/C catalyst after AAT [34].

Table 7
Thermal properties.

Description	Value	Ref.
Thermal conductivity of the hydrogen, k_{H_2}	0.2 W/m•K	[39]
Thermal conductivity of the oxygen, k_{O_2}	0.0296 W/m•K	[39]
Thermal conductivity of the water vapor, k_{H_2O}	0.0237 W/m•K	[39]
Thermal conductivity of the nitrogen, k_{N_2}	0.0293 W/m•K	[39]
Thermal conductivity of membrane, k_{mem}	0.95 W/m•K	[41]
Thermal conductivity of GDL/CL, k_{GDL} , k_{CL}	1.2 / 1.5 W/m•K	[47]
Thermal conductivity BP, k_{BP}	20 W/m•K	[38]

ANSYS FLUENT ver. 19 (ANSYS, Inc., USA) was used to numerically implement the PEMFC model. In particular, user-defined functions were used. The convergence criterion for equation residuals was set to 10^{-8} , and the number of grid points required to sufficiently resolve spatial gradients in typical computational domains was 1660,000 (i.e., approximately $83 \times 20 \times 1000$ in the x , y , and z directions, respectively).

3. Results and discussion

By utilizing the PDF approach for PEMFC modeling, we investigated the impact of the Pt particle size distribution on simulation results of PEMFCs. Fig. 3, reproduced from Liu et al. [34], shows

Table 8
Boundary conditions.

Description	Expression	
Anode inlet velocity	$u_{in,an} = \frac{\xi_{an}}{C_{H_2}} \frac{A_{mem}}{A_{an,chan}}$	(35)
Cathode inlet velocity	$u_{in,ca} = \frac{\xi_{ca}}{C_{O_2}} \frac{A_{mem}}{A_{ca,chan}}$	(36)
Electric potential on the anode outer BP surface	$\phi_s = 0$	(37)
Electric potential on the cathode outer BP surface	$\phi_s = V_{cell} \text{ or } \sigma \nabla \phi_s = I$	(38)

histograms of Pt particle size distributions for Pt/C and Pt/TiO₂/C catalyst structures wherein different degrees of Pt particle growth in Pt/C and Pt/TiO₂/C catalysts can be clearly observed. The mean Pt particle radius of around 2.65 nm for the pristine Pt/C catalyst (Fig. 3a) increased to around 13.25 nm after the accelerated aging test (AAT) (Fig. 3b). Furthermore, in Fig. 3c and d, owing to

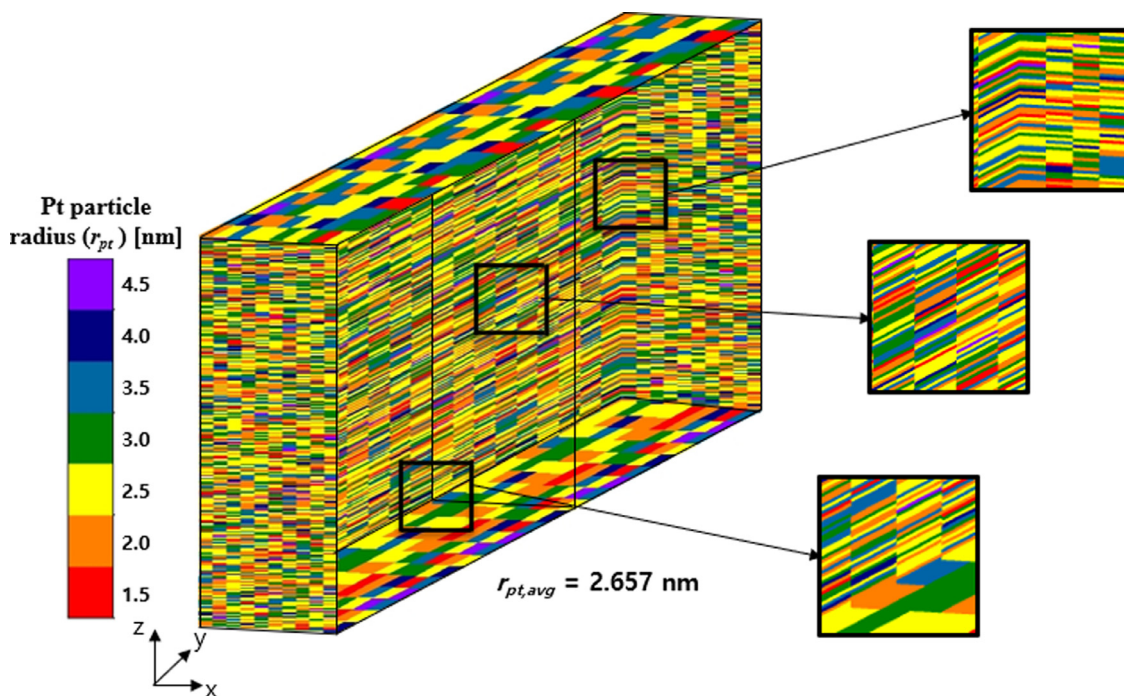


Fig. 4. Contours of Pt particle distributions for the pristine Pt/C CL in Fig. 3a.

the effect of the semiconductive material TiO_2 , the Pt/ TiO_2 /C catalyst shows stable performance in terms of Pt particle growth compared with the Pt/C catalyst, with the mean particle radius increasing from 3.65 nm to 4.6 nm. The histograms in Fig. 3 show that a CL domain with various Pt particle sizes was realized with a sufficient number of elements, with the Pt particle distribution being similar to that observed in experimental data. For instance, Fig. 4 shows contours of Pt particle distributions in the CL domain for the pristine Pt/C CL in Fig. 3a. Clearly, the Pt particle size range 2–2.5 nm is dominant. Consequently, the mean Pt particle size in the computational CL domain was set to about $r_{pt,avg} = 2.657$ nm, sufficiently close to the average Pt size in the experimental data ($r_{pt,avg} = 2.65$ nm) in Fig. 3a. This implies that the statistical approach to generating a random distribution of Pt particles successfully captures an experimentally constructed CL. Since key CL properties such as ECSA and oxygen transport resistance are strong functions of the Pt particle radius r_{pt} , local Pt particle size distributions can influence the electrochemical reactions and oxygen transport behavior. In particular, ECSA may directly affect the cell performance, which implies that a 3D PEMFC model with various Pt particle sizes in the CL can more accurately estimate the impact of micro-CL properties on the cell performance.

In Fig. 5, the PEMFC model combined with the PDF-based new microscale CL model was first experimentally validated against experimental data obtained for various CL designs and degradation conditions [34]; in the experiment, AATs were performed by employing the potential sweeping test in the range of 0.6–1.2 V for 3000 cycles. To assess the impact of Pt particle size distributions more clearly, multiscale PEMFC simulations were conducted with the average Pt particle size ($r_{pt,avg}$) as well as with experimentally observed variation of the Pt particle size. In general, the simulation results agreed well with experimental data obtained for Pt/C and Pt/ TiO_2 /C catalyst structures before the AAT and after the AAT. For the Pt/C catalyst before the AAT and the Pt/ TiO_2 /C catalyst before and after the AAT (Fig. 5a, 5c, and 5d, respectively), when

the variation in Pt particle size was small, the performance difference between the use of the average Pt size ($r_{pt,avg}$) and the actual variation of the Pt size was small and negligible. By contrast, the variation of the Pt particle size was large for Pt/C after the AAT, and the impacts of considering the actual range of Pt particle sizes and their spatial variation became substantial. Interestingly, Fig. 5b indicates that the assumption of the average Pt radius as $r_{pt,avg} = 13.25$ nm underestimated the cell performance, while excellent agreement with experimental data was achieved by considering the actual Pt particle size distribution ($2.5 \text{ nm} \leq r_{pt} \leq 37.5 \text{ nm}$ in Fig. 3b). The higher performance obtained with $2.5 \text{ nm} \leq r_{pt} \leq 37.5 \text{ nm}$ compared with $r_{pt,avg} = 13.25 \text{ nm}$ could imply that the number of Pt particles smaller than the average size ($r_{pt} \leq r_{pt,avg} = 13.25 \text{ nm}$) was sufficient under the high Pt loading condition ($L_{Pt} = 0.4 \text{ mg/cm}^2$) and most of them participated in the ORR, while Pt particles larger than $r_{pt,avg} = 13.25 \text{ nm}$ may not have participated in the reaction.

Fig. 6 shows contours of current density distributions in the membrane at $I = 1.5 \text{ A/cm}^2$. When the random spatial variation of different Pt particle sizes was considered, local fluctuations appeared in the current density profiles, clearly indicating nonuniform oxygen transport in the microscale Pt/C and Pt/ TiO_2 /C particle domains. In Fig. 6b, a higher degree of current density fluctuation is clearly observed for a wider range of Pt particle size distributions for the Pt/ TiO_2 /C catalyst after the AAT. The total oxygen transport resistance through the ionomer and water films, Ω_T , is plotted in Fig. 7. Since Ω_T is proportional to the effective oxygen permeation length through the ionomer film and water film to reach a Pt particle, denoted by δ_o^{eff} and δ_w^{eff} in Eq. (23), a spatial inhomogeneous pattern is observed for the random distribution of Pt particle sizes.

The aforementioned analysis was based on a high Pt loading of 0.4 mg/cm^2 , at which the Pt particle size distribution may not significantly influence the cell performance and oxygen transport in the microscale CL structure. Therefore, additional simulations were

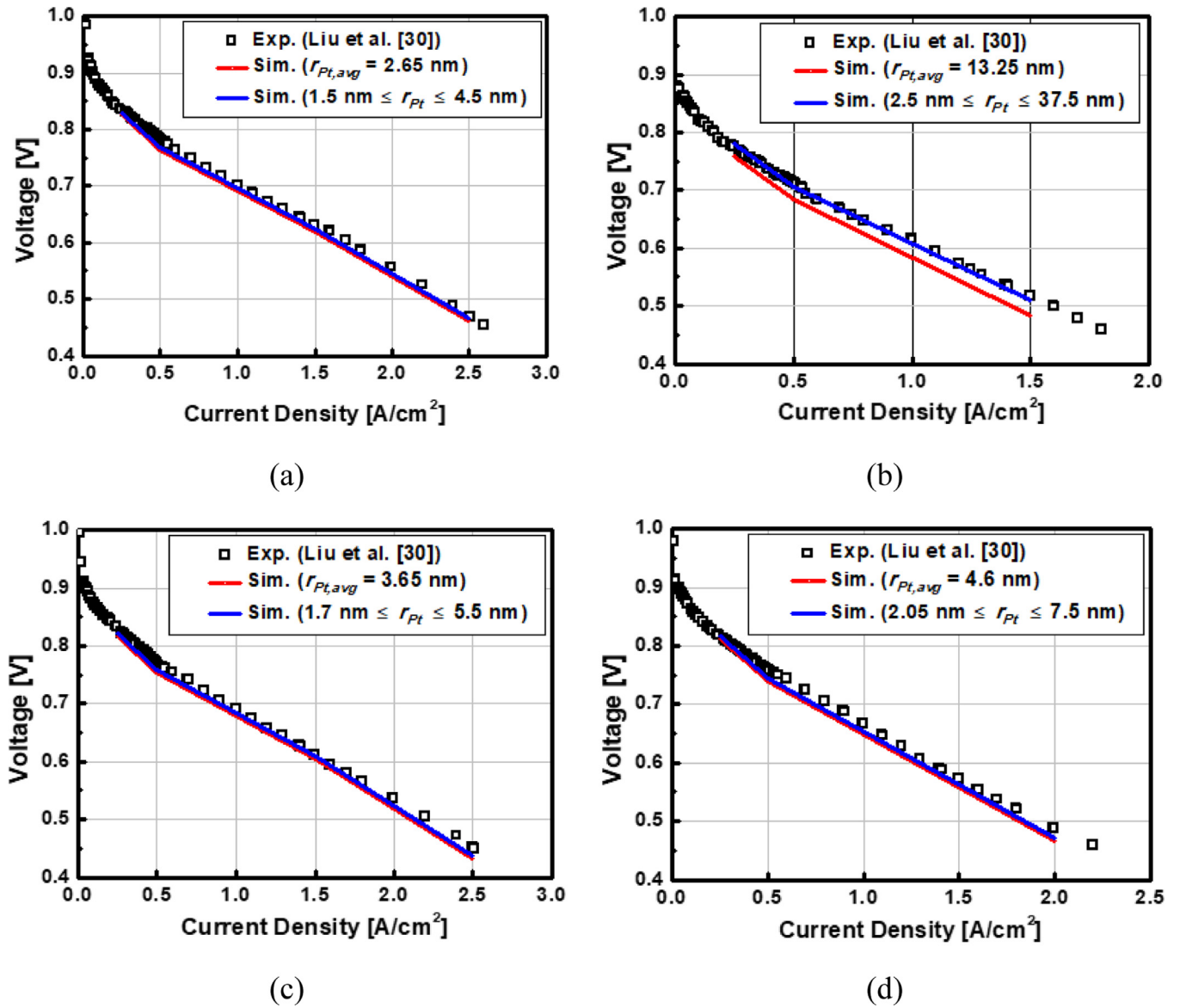


Fig. 5. Comparison of simulated polarization curves with the experimental data measured by Liu et al. [34]: (a) Pt/C before AAT, (b) Pt/C after AAT, (c) Pt/TiO₂/C before AAT, and (d) Pt/TiO₂/C after AAT. The Pt loading for the simulations and experiments is $L_{Pt} = 0.4$ mg/cm².

performed at a lower Pt loading of 0.05 mg/cm², and the key results obtained are plotted in Fig. 8. Clearly, the degree of nonuniformity in the current density, the oxygen concentration, and oxygen transport resistance distributions increased when the Pt particle size variation was considered, resulting in lower cell performance. Since the simulation results in Fig. 8 are for the pristine Pt/C catalyst before the AAT with a narrow range of Pt particle sizes ($1.5 \text{ nm} \leq r_{Pt} \leq 4.5$ nm), apparently, the consideration of a nonuniform Pt size distribution in PEMFC simulations is particularly more important when the CL is degraded at low Pt loading values, which results in nonuniform Pt particle growth followed by a wider range of Pt particle size distributions.

4. Conclusions

In this study, we applied a statistical PDF approach to a multi-scale PEMFC model to randomly distribute Pt particles with differ-

ent sizes in a computational domain under the assumption that the Pt particle size distribution followed a normal distribution curve. The combination of the PEMFC model and PDF approach was successfully validated against experimental data in which two different CL designs with Pt/C and Pt/TiO₂/C were degraded by the AAT, resulting in a wide range of Pt particle size distributions, from several nanometers to tens of nanometers. The simulation results indicate that the consideration of different Pt particle sizes is particularly more important when Pt loading is low and the degradation of the CL driven by agglomeration and coalescence of Pt particles is substantial. Furthermore, a fluctuation pattern representing an inhomogeneous Pt particle size distribution in the CL was observed in the multidimensional contours of the current density, oxygen concentration, oxygen transport resistance distribution.

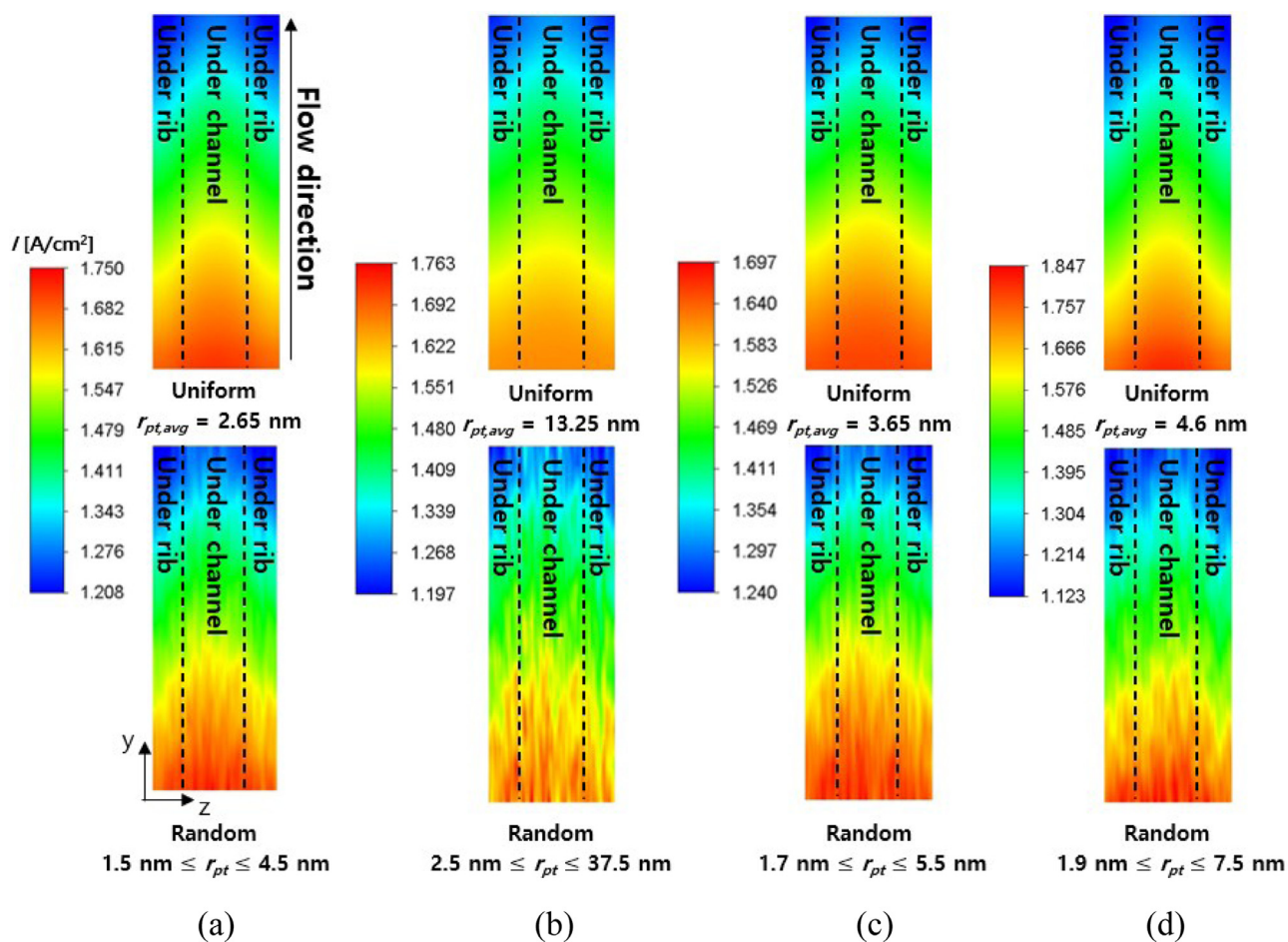


Fig. 6. Contours of current density distributions in the membrane at 1.5 A/cm² with $L_{Pt} = 0.4 \text{ mg/cm}^2$: (a) Pt/C catalyst before AAT, (b) Pt/C catalyst after AAT, (c) Pt/TiO₂/C catalyst before AAT, and (d) Pt/TiO₂/C catalyst after AAT. The use of $r_{pt,avg}$ in the upper figures represents simulations with the average Pt particle size while the actual variation of Pt particle size (r_{pt}) was considered for the simulations in the lower figures.

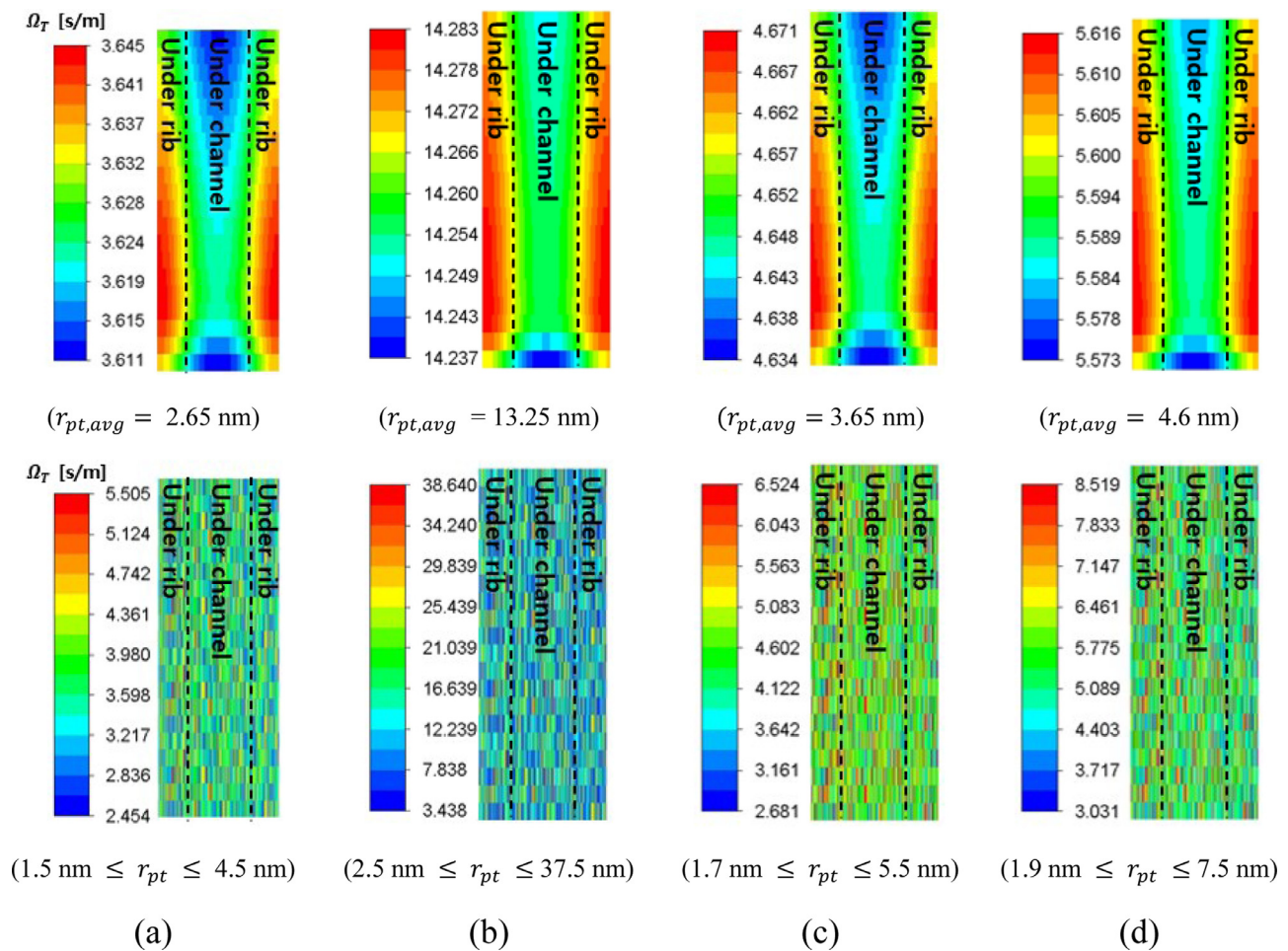


Fig. 7. Contours of oxygen transport resistance distributions at 1.5 A/cm² with $L_{Pt} = 0.4$ mg/cm²: (a) Pt/C catalyst before AAT, (b) Pt/C catalyst after AAT, (c) Pt/TiO₂/C catalyst before AAT, and (d) Pt/TiO₂/C catalyst after AAT. The use of $r_{pt,avg}$ in the upper figures represents simulations with the average Pt particle size while the actual variation of Pt particle size (r_{pt}) was considered for the simulations in the lower figures.

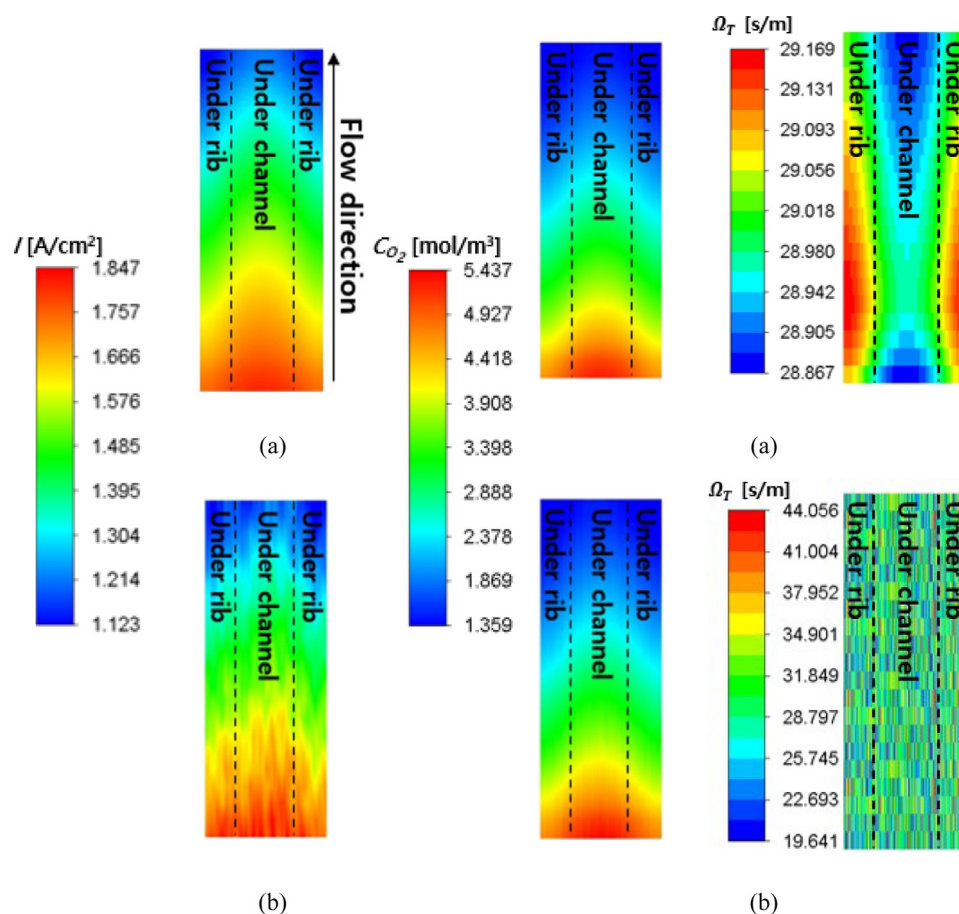


Fig. 8. Contours of current density distribution in the membrane, I [A/cm^2], oxygen concentration C_{O_2} [mol/m^3], and oxygen transport resistance Ω_T [s/m] distribution on Pt particles located in the middle plane of the cathode CL at $1.5 \text{ A}/\text{cm}^2$ and $L_{Pt} = 0.05 \text{ mg}/\text{cm}^2$: (a) average Pt particle size $r_{pt,avg} = 2.65 \text{ nm}$ and (b) actual range of Pt particle size, $1.5 \text{ nm} \leq r_{pt} \leq 4.5 \text{ nm}$.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Credit authorship contribution statement

Jaeyoo Choi: Writing – original draft, Formal analysis, Data curation. **Eunsoo Kim:** Writing – original draft, Formal analysis. **Yohan Cha:** Formal analysis, Data curation. **Masoom Ghasemi:** Writing – review & editing. **Hyunchul Ju:** Conceptualization, Writing – review & editing, Supervision.

Acknowledgment

This study was supported by the Korea Evaluation Institute of Industrial Technology (KEIT) and the Ministry of Trade, Industry and Energy (MOTIE), Republic of Korea (No. 20012121). We would like to thank TAESUNG S&E, Inc., Seoul, Republic of Korea, for providing technical support for the Ansys Fluent software.

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